

End-to-End Animation Revamp Guide

How to take `/w/ananya-weds-arjun` (and the staff pages) from “nicely animated” to *alive* — which libraries to use in 2026, exactly what to build on each page, in what order, and how to verify it without breaking the demo. All library versions, licences and browser-support facts below were verified against npm and vendor docs on 12 July 2026.

1 • Where the site is today

The current animation layer is deliberately dependency-free: CSS keyframes for hero entrances (`.rise`, letter-stagger `.ltr`, foil shimmer), an IntersectionObserver in `sections/ScrollFx.tsx` driving `.reveal` / `.draw` / `.rule-grow` scroll reveals, CSS-only petals, marquee, scallop band edge, and the new “How we met” chapter carousel. It works in every browser, survives reduced-motion, and costs 0 kB of JS beyond one tiny client component.

What it *can't* do — and what this revamp adds: **pinned scroll scenes** (a section that holds while its content scrubs), **scrub-linked timelines** (animation progress tied to scroll position, not just “entered viewport”), **masked line-by-line text reveals**, **FLIP layout transitions** (gallery → lightbox), and **celebration moments** (RSVP confetti). Those genuinely need a library.

2 • The 2026 library landscape (verified)

LIBRARY	VERSION / LICENCE	SIZE (GZIP)	USE IT FOR	DOCS
GSAP + ScrollTrigger + SplitText	3.15 — 100% free since 3.13 (Webflow acquisition); all former Club plugins (SplitText, DrawSVG, MorphSVG, Flip...) now ship free in the <code>gsap</code> npm package	core ~27 kB, +13 ScrollTrigger, +7 SplitText	Scroll storytelling: pinning, scrubbing, snap; masked text reveals; SVG draw; the gold standard	gsap.com/docs/v3
anime.js v4	4.5 (MIT) — full ESM rewrite Apr 2025	~39 kB full, tree-shakes far smaller	SVG draw/morph (<code>svg.createDrawable</code>), springs, draggable, built-in text splitter; <code>onScroll</code> scrubs but has no pin/snap	animejs.com/documentation
Motion (ex-Framer Motion)	12.42 (MIT), import from <code>motion/react</code>	34 kB → ~6 kB via <code>LazyMotion</code>	Declarative React micro-interactions, <code>AnimatePresence</code> exits, <code>layout</code> morphs, light <code>useScroll</code> parallax	motion.dev/docs/react
Lenis	1.3 (MIT) — old <code>@studio-freight/*</code> packages deprecated; React adapter is <code>lenis/react</code>	5.2 kB	Buttery smooth scroll. Keeps <code>position:sticky</code> and anchors working; CSS scroll-snap needs its plugin	lenis.dev
Embla	embla-carousel-react 8.6 (MIT) — the engine inside shadcn/ui's carousel	7.1 kB	Headless carousel (swipe physics, free-drag). Upgrade path for the “How we met” carousel	embla-carousel.com
canvas-confetti	1.9 (ISC), zero deps	~6 kB	The RSVP-submitted celebration. Dynamic-import on submit only	npm: canvas-confetti
Native CSS	scroll-driven animations: Chrome 115+, Safari 26+, Firefox still flagged · <code>@starting-style</code> : Baseline · View Transitions: same-doc everywhere, cross-doc Chrome/Safari	0 kB	Progressive enhancement only — the site already treats it that way (our old <code>view()</code> timeline bug is why)	MDN

Recommended stack for this project: *GSAP (ScrollTrigger + SplitText + DrawSVG) as the scroll-narrative engine*, via the official `@gsap/react useGSAP()` hook — SSR-safe, auto-cleanup, made for the App Router. Add **Embla** to harden the story carousel, **canvas-confetti** for RSVP success, and optionally **Lenis** for scroll feel (decide once, early — see §4). Skip Motion for now: it shines on app-like UI, and the staff portal deliberately stays quiet. anime.js v4 is the lighter alternative if you'd rather avoid pinned scenes entirely — everything in §5 marked *[pin]* is GSAP-only.

3 • Ground rules (read before writing any animation)

- **Keep the CSS layer as the fallback.** The IntersectionObserver reveals stay; GSAP progressively takes over sections it owns (add `data-gsap` and skip those in ScrollFx) — never leave content at opacity 0 if JS fails: GSAP sets, CSS defaults to visible.
- **Reduced motion is law.** Wrap every effect in `gsap.matchMedia().add("(prefers-reduced-motion: no-preference)", ...)`; the CSS layer already guards itself.
- **Theme tokens only.** Animate `var(--accent)`, `var(--hairline)` etc. All three themes must inherit every effect free of charge, like the existing system does.
- **Transforms + opacity only** on scroll-linked tweens (compositor-friendly). Never animate layout properties in a scrub.
- **Client-component boundaries:** animation code lives in `"use client"` leaf components (like `ScrollFx.tsx` today); sections stay server components and pass refs down.
- **No CLS:** pinned sections reserve their height (ScrollTrigger handles this with `pinSpacing`); the countdown already reserves cell height — keep that pattern.
- **Test gates:** after each page, run `npm test` and `npx playwright test` (stop the dev server first — PGLite is single-process), plus a 390 px pass in the browser.

4 • Setup (once)

```
# in ~/projects/wedding-mvp, on a branch
git checkout -b feat/animation-revamp
npm i gsap @gsap/react embla-carousel-react embla-carousel-autoplay canvas-confetti
npm i lenis # optional – decide now, not later
```

The one hook pattern used everywhere (SSR-safe, self-cleaning):

```
"use client";
import gsap from "gsap";
import { ScrollTrigger } from "gsap/ScrollTrigger";
import { SplitText } from "gsap/SplitText";
import { useGSAP } from "@gsap/react";
gsap.registerPlugin(ScrollTrigger, SplitText, useGSAP);

export function HeroFx({ scope }: { scope: React.RefObject<HTMLElement> }) {
  useGSAP(() => {
    const mm = gsap.matchMedia();
    mm.add("(prefers-reduced-motion: no-preference)", () => {
      // tweens here – auto-reverted on unmount, scoped to `scope`
    });
  }, { scope });
  return null;
}
```

Lenis decision: if you add it, mount one provider in the public site layout and wire it to ScrollTrigger — and then *don't* also lean on CSS `animation-timeline` anywhere (double smoothing feels wrong). Skip Lenis entirely if native scroll feel is fine on the demo laptop; it is the most reversible choice here.

```
"use client";
import { ReactLenis, useLenis } from "lenis/react";
import { ScrollTrigger } from "gsap/ScrollTrigger";
export function SmoothScroll({ children }) {
  useLenis(ScrollTrigger.update);
  return <ReactLenis root options={{ lerp: 0.12 }}>{children}</ReactLenis>;
}
```

5 • Page-by-page revamp plan

5.1 Hero — the first five seconds

- **Masked line reveal for the names** (replaces per-letter CSS stagger): SplitText with `type: "chars, lines"`, `mask: "lines"` — chars rise out of a clipped line, the exact effect on luxury invite sites. ~8 lines of code now that SplitText is free.
- **Scrub-linked hero exit** [pin]: as you scroll away, names drift up at 0.8×, watermark ampersand at 0.3×, petals accelerate — one timeline with `scrollTrigger: { trigger, start: "top top", end: "bottom top", scrub: true }`. Replaces the single-value parallax in ScrollFx.
- **Countdown flip**: on each second, tween the changed digit `yPercent: -100` inside an `overflow:hidden` slot (odometer feel, serif digits). Pure GSAP micro-tween in the existing `Countdown.tsx`.
- Keep: petals (CSS is perfect for ambience), foil ampersand, scroll cue.

5.2 “How we met” story — chapter scrub

- **Option A (bigger wow) [pin]:** replace the timed carousel with a pinned scene — section pins for ~ 2.5 viewport-heights, chapters crossfade/slide as the visitor scrolls, progress bar maps to scroll progress (`snap: 1/(chapters-1)` gives it that chaptered feel). Scroll becomes the navigation; dots/arrows stay as accessible alternates.
- **Option B (lower risk):** keep auto-advance but move the engine to Embla (+ `embla-carousel-autoplay`) — real swipe physics on mobile, and the current hand-rolled gesture/focus logic shrinks substantially.
- Ornament dividers: swap the CSS dash trick for DrawSVG on scroll — smoother, and it can draw *and* fade the fill in one timeline.

5.3 Event timeline — the itinerary as a journey

- **Scrubbed progress thread:** a vertical line in the number column that fills top→bottom with scroll (`scaleY scrub`), each event node igniting to `var(--accent)` as the thread passes — the classic wedding-itinerary treatment, and ScrollTrigger’s `containerAnimation` -free bread and butter.
- **Row choreography:** `ScrollTrigger.batch()` the rows — number counts up (`textContent snap`), title rises with a SplitText word-stagger, venue block follows 0.1s later.
- Dress-code chips: tiny hover spring (scale 1.04, border → accent) — `contextSafe` handlers.

5.4 Gallery — plates that open

- **Flip lightbox:** click a plate → it FLIPS (GSAP Flip plugin) from its grid cell to a centered, enlarged card over a dimmed backdrop; ESC/click FLIPS it home. Works today with the SVG plates and inherits real photos after the haldi with zero changes.
- **Reveal upgrade:** `ScrollTrigger.batch` with a small random rotation ($-1.5^\circ \dots 1.5^\circ$) per plate so the wall settles like hung frames.

5.5 RSVP band + RSVP page — the payoff

- **Band entrance:** SplitText line-mask on the big headline as the band scrolls in; scallop edge is already the divider — no change.
- **Confetti on submit** (on `/rsvp/[token]`): after a successful attending response — `const confetti = (await import("canvas-confetti")).default; confetti({ particleCount: 90, spread: 70, colors: [accent, accent2, "#fff"] })` — read the two accents from `getComputedStyle` so even the confetti is themed. Skip under reduced motion.
- **Stepper micro-feel:** headcount +/- buttons get a 1.06 scale spring; per-event cards get `@starting-style` entry fades (zero JS, Baseline support).

5.6 Staff portal (signin/admin/committee/couple) — restraint

- These pages should feel like a calm tool, not the wedding. Limit to: number **count-ups** on stat cards (one 0.6s tween on mount), **row stagger** on first table paint (30ms/row, opacity+4px rise), and `@starting-style` fades for modals (family editor) and the sent-link confirmation. No scroll effects.

6 • Execution order (end to end)

1. **0 Branch + install** (§4). Decide Lenis now. Commit the lockfile.
2. **1 Shared fx utilities:** `src/fx/` with the registered-plugins module and a `<SectionFx>` wrapper; mark GSAP-owned sections `data-gsap` and make `ScrollFx.tsx` skip them. *Gate: site still renders with JS disabled.*
3. **2 Hero** (§5.1). *Gate: LCP unchanged $\pm 10\%$, 390px clean, reduced-motion shows static hero.*
4. **3 Story** (§5.2 — pick A or B). *Gate: keyboard can still drive chapters; e2e passes.*
5. **4 Timeline + Gallery** (§5.3–5.4). *Gate: no CLS from pinning; Flip returns cleanly on ESC.*
6. **5 RSVP moments** (§5.5). *Gate: e2e RSVP flow green — confetti must not block submission; it fires after the server action resolves.*
7. **6 Staff polish** (§5.6). *Gate: unit tests green.*
8. **7 Full pass:** stop dev server → `npm test` → `npx playwright test` → `npm run build` → reseed → restart dev → click through all three themes desktop + 390px. Then merge and (post-Task-16) `vercel --prod`.

Each step is a commit; each gate is cheap insurance that the demo never regresses. Total effort estimate: steps 1–2 one evening, 3–5 a weekend, 6–7 an evening.

7 • Resources

- **GSAP docs** — gsap.com/docs/v3 · React guide: gsap.com/resources/React · live demos: demos.gsap.com
- **GSAP-is-free announcement** — webflow.com/blog/gsap-becomes-free · licence: gsap.com/standard-license
- **anime.js v4** — animejs.com/documentation (`createTimeline`, `svg.createDrawable`, `createScope` for React)
- **Motion for React** — motion.dev/docs/react (`AnimatePresence`, `layout`, `useScroll`)
- **Lenis** — lenis.dev · github.com/darkroomengineering/lenis (`ScrollTrigger` wiring in README)
- **Embla** — embla-carousel.com (also the engine behind shadcn/ui's carousel)
- **Codrops** — tympanus.net/codrops/hub · “From SplitText to MorphSVG: 5 creative demos using free GSAP plugins” (May 2025) · tympanus.net/codrops/tag/scroll
- **Native CSS scroll-driven animations** — webkit.org/blog/17101 (Safari guide) · caniuse.com/mdn-css_properties_animation-timeline_scroll (Firefox still flagged — keep it enhancement-only)
- **Inspiration** — godly.website · awwwards.com · minimal.gallery · blissandbone.com (editorial benchmark) · ravishkapoor.com / puneetguptainvitations.in (luxury Indian invite restraint)